

Algorithm Trading through Application Programming Interface (API) in Indian Stock Market

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Abstract— Algorithmic Trading (AT) is a process to execute orders using automated and preprogrammed trading instruction by using Price, Timing and Volume as a variable. Algorithms are designed by approved vendors which work on past data for achieving set objectives. A human emotion free decision taken through ATs. AT came in India in 2008, initially institutional investors started trading with AT, retail investors adopted AT in last couple of years. Indian stock market regulator SEBI trying to protect investors from harms of AT. In recent past SEBI has also released a consultation paper on algorithmic trading by retail investors dated on Dec 19, 2021. This study is examining the current status and strategic analysis of algorithm trading w.r.t to Application Programming (API) algorithm trading in Indian stock market. This study will be useful for all investors using AT or planning to use AT. It will also be helpful for policy makers to frame futuristic policies on algorithmic trading.

Keywords— Algorithmic Trading (AT), Application Programming Interface (API)

I. INTRODUCTION

Stock trading now dominated by mechanized trading due to automation of industry and trading decisions are heavy influence by data driven strategies. If ATs are designed by a vendor who want to take the advances of low technical competency of its users, these ATs are may be used for market making activities. Algorithmic Trading (AT) is just another way of conducting trading activities between two parties using computer programs whose goal is to conduct trade in the most efficient way possible [1]. Algorithm trading is a computer program which analyze the market data and generate buy and sells indicators based on pre decided strategies for the exchanges. According to the business wire 8.8 trillions of financial assets worldwide are controlled by algorithm trading till 2016 which can grow by 8.7% till 2025, as per JP Morgan 60% equity now and traded on algorithm trading. Algorithm Trading (AT) came in India in 2008 initially few investors were using it, retail traders started using advanced AT from just five years back [2]. In the last few decades' financial markets have seen a lot of changes when it comes to trading securities where the various stages of trading activity have shifted from manual to electronic systems [3]. AT are getting attentions of investors due to its unique advantages like Trend Identification: AT process larger chunk of data which helps in finding major trends which prevails in stock market. Trends are useful for developing multiple strategies (buy/sell) even in a single stock. Data Neutral Strategies: it's very difficult to find out delta change of derivative price due change on its underlying asset in manual trading. Through AT we can apply the multiple positions to balance negative delta. Trading

Minus Human Emotions: human emotions are completely eliminated in AT, machines are not emotionally attached to any stocks, it also avoids the human errors like mistakes due to the physical limitations during the trade time. Positioning Sizing: AT balance the trading positions by applying multiple positions at a time. Stop Loss Modifications, Scalping and many other advantages are associated with algorithmic trading.

II. PRESENT TRADING IN INDIAN STOCK EXCHANGES

First, Colocation Mode of Trading: this mode is most common mode of trading in either BSE (36.16% of total as on sept 22) or NSE (33.68% of total in sept 22). This is a facility where dedicated spaces in the exchange building, right next to the exchange servers, where high-frequency and algo traders can place their systems or programs. Due to this OPG securities and Money Life exposed for a big scam in scam in NSE.

Mobile Trading: Mobile trading (MT) refers to the use of wireless technology in securities trading and allows investors to use their smartphones to trade. Mobile trading is used in very good percentage in both the trading platform [4]. MT can be performed through using of mobile app of a member trader, there are so many mobile app of various brokers which are popular in market like Zerodha Kite App, ICICI direct Markets, m.Stock, 5Paisa Mobile App, Aliceblue, Angel Broking Mobile App, Groww App, Upstox Pro App, IIFL Markets App, HDFC Securities App

Inter Based Trading (IBT): IBT is another popular mode of trading in both the exchanges (NSE: 19.62 and BSE: 7.25 percent as on Sept 2022), in IBT exchanges are offering IBT servers access to their members and their clients to access from anywhere. NSE is leading in this mode because it has more equipped technical infrastructure than BSE. The market regulator SEBI reviewed IBT and has issued several guidelines on Jan 31, 2000, those are capturing of IP address of that IBT transaction, end to end encryptions requirement, automatic logout of client login if there is no transaction during a particular tenure etc. [5].

Direct Market Access (DMA): DMA is a mechanism where client/investors are accessing the market directly by using CTCL (commuter to computer link) software of trading member. This mode of trading is very minimal in both the exchanges.

Smart Order Routing (SOR): SEBI permitted SOR in August 27, 2010 in this the broker trading engines are permitted to systematically choose the based on certain factors related to

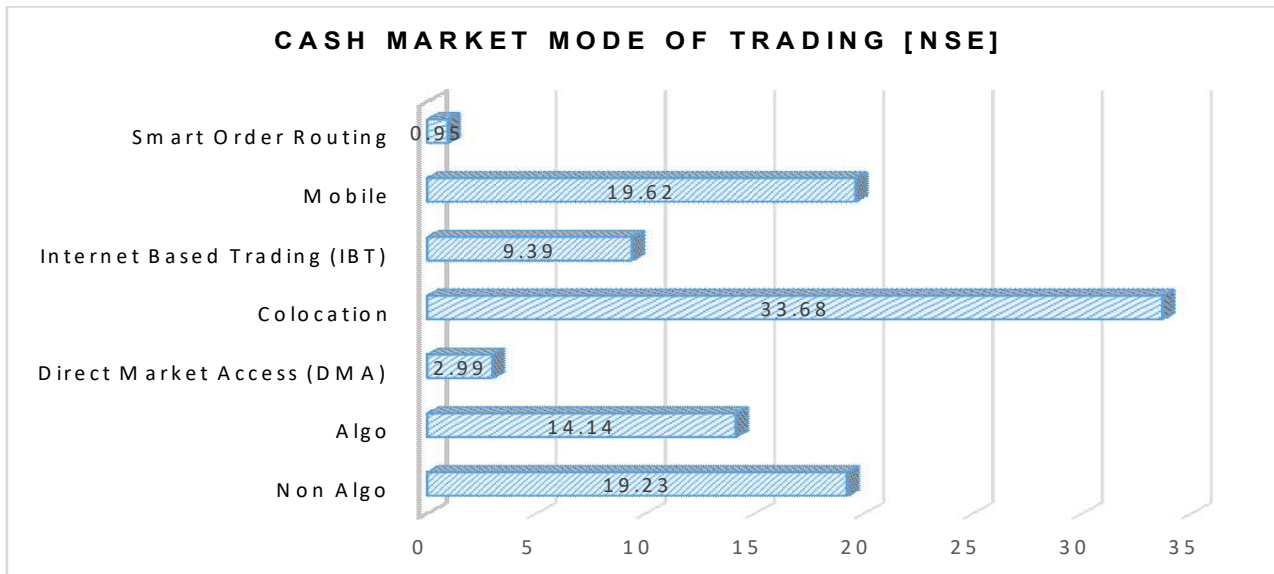


Fig: 1(a) Cash Market Mode of Trading in NSE

the order execution like price, cost, speed, size, nature etc. Data says that SOR is not widely traded mode in both exchanges.

III. APPLICATION PROGRAMING INTERFACE (API) ALGORITHM TRADING

Application Programming Interface allow a trader to code a computer program in the terminal of trader or API allow to use trading platform of any other for trading. In case of trade algorithm developed by a trader and algorithm is plug in their own system API is providing real time information related to price and volume. API in trading is used by two ways, first is trader is using third party applications in this case trader need broker API for real time data related to price and volume and the second is trader is using their own trading algorithm but gain need for real time data. Use of API for trader is easily understood by following process (fig:2)

Trading API: These API allows traders to customize trading platforms by using third party software. These may be offered by broker with coded algorithm on brokers trading platforms. TAPI (Trading API) provides various solutions to the traders like Real Time Order Placement, Track Order Executions, Portfolio Management etc. API Trading is a low-cost trading methods to build an algorithm trading system, in India API are offered free of cost or on charge basis [6]. A detail description on dimensions of trading API is explained in fig.3. APIs has not only systematized algorithm trading but also has increased the size of algo trading [7]

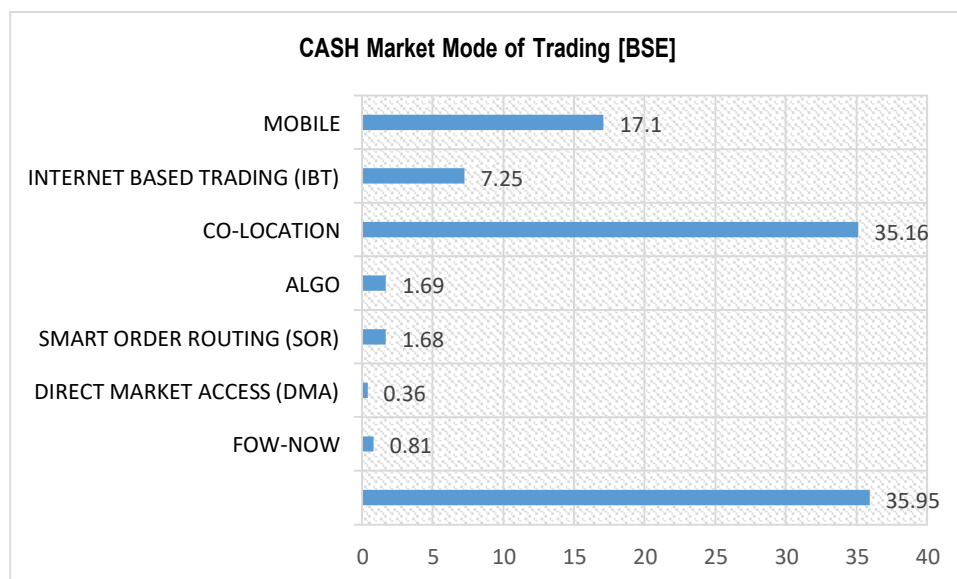


Fig: 1(b) Cash Market Mode of Trading in NSE

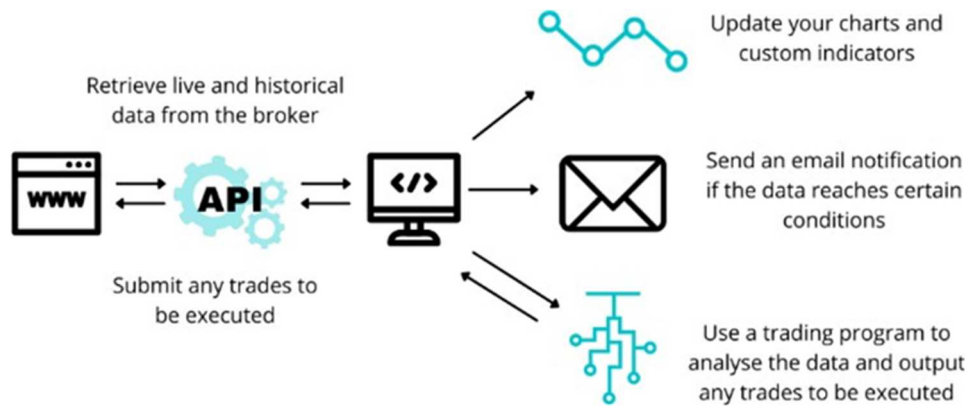


Fig: 2 API: Functioning

IV. APPLICATION PROGRAMING INTERFACE (API) ALGORITHM TRADING AND ITS REGULATION IN INDIA

After Regulator view of algorithm trading is different, one is high frequency traders and AT are viewed market- maker if they act on interest of some companies this may lead a major disruption on market. Another view is although algorithm trading provides liquidity that may lead some disruptive practices like herding, predatory trading, order manipulation, or burden the stock exchange with excessive messages. In this context SEBI has also released a consultation paper on algorithmic trading by retail investors dated on Dec 19, 2021, in this paper SEBI highlight the concerns over the rise of unregulated algorithm2 platforms. SEBI proposed more responsibilities to brokers like all orders generated through

API should be treated as algo order, in this case unique id should be approved by exchange is to be given to those orders, and the used algorithm strategies will also be audited. Brokers allowing API access to clients will create an entity which will manage the server from where these orders are generated. The another important issue in the consultation paper is related to the algo broker will provide their in-house or approved third party algorithm strategies.

V. CONCLUSION

AT can empowered stock markets through its several unique competitiveness like improvisations of liquidity because, AT is supplying liquidity by placing limit orders in the market and hence AT might be improving the overall market liquidity as well. The other uniqueness is minimal volatility due its efficacy in technology [8] Trading can

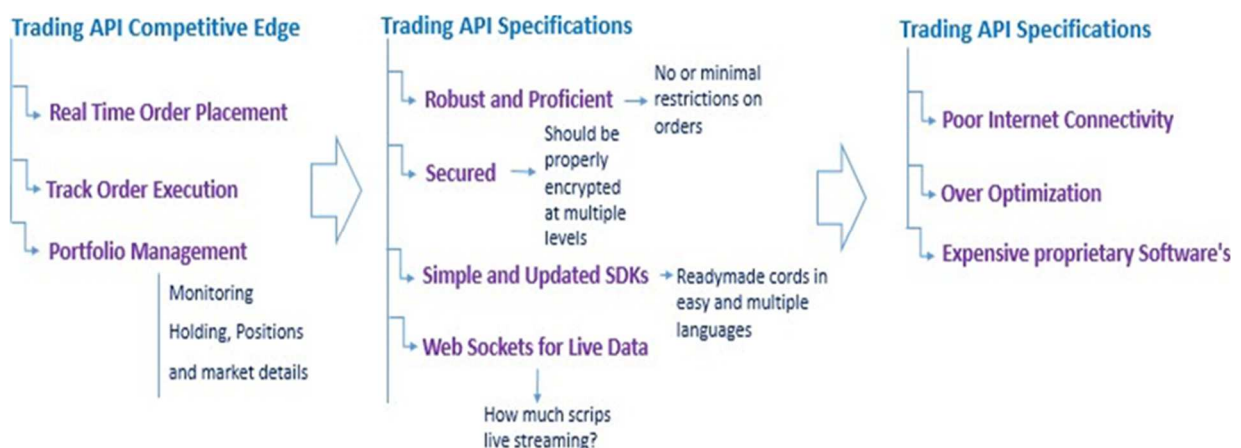


Fig:3 Trading APIs Dimensions in Indian Context

change their trading style by using API and can make it more professional. API Algo trading has its unique presence in stock market, growing participations of retail investors and their technology based preference of trading is the strength for the API also trading. The low level of technical literacy and improper regulatory framework of API Algo Trading is the weakness. Technology based trading trends in stock market are the opportunities for the API AlgoTrading and emergence of unregulated algorithm are the main threats for API Algo Trading. It can have concluded as APIAT is the future opportunity for Indian stock market which can be empowered through investors technology awareness and structured regulatory framework.

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